

Kerem Yildirim

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Munich, Germany



EDUCATION

- **M.Sc. Informatics**

Technical University of Munich

Thesis Topic: Object detection and scene completion from a single RGB-D frame

Grade Average: 1.7

October 2019 - September 2022, Germany

- **B.Sc. Computer Science and Engineering**

Sabanci University

Thesis Topic: Plant Disease Identification

Grade Average: 1.4

September 2015- June 2019, Turkey

EXPERIENCE

- **IT Consultant - Netlight**

Specializing in Cloud/DevOps, MLOps

October 2021 - Present, Germany

- **Machine Learning Engineer - Isarsoft**

Trained and deployed multiple object detection solutions to different infrastructures ranging from cloud to edge devices.

October 2021 - Present, Germany

- **Computer Vision Engineer - Research Neutron Source FRM II**

Developed a system for high speed image segmentation using C++ and CUDA, enabling the processing of large datasets.

October 2020 - October 2021, Germany

- **Computer Vision Intern - WARP**

Designed and implemented a 3D object detection, instance segmentation and tracking system on LiDAR point clouds using Python and ROS to be used in an autonomous vehicle for in-campus transportation.

April 2020 - September 2020, Germany

LANGUAGES

- Turkish - Native
- English - C1
- German - B1

SKILLS

- C/C++, Python, CUDA, MATLAB, $\LaTeX$
- ROS, PyTorch, Tensorflow, OpenCV, OpenMP
- Linux, CMake, Git, Docker, AWS, Azure

PROJECTS

- **Monte Carlo Dropout for Object Detection on Point Clouds**

Improved state of the art object detection performance on point clouds by leveraging uncertainty.

Python, PyTorch

- **Indirect Visual Inertial Odometry with Optical Flow**

Extended the classical odometry pipeline by eliminating the descriptor matching and integrating frame to frame optical flow for more robust feature tracking.

C++, CMake, Eigen, Sophus, Ceres

- **KinectFusion**

Implemented KinectFusion algorithm by Newcombe et al. for real-time reconstruction of real world 3D scenes from depth images.

CUDA, C++, CMake, Eigen, Ceres

- **Divergence-Free Shape Correspondences**

Built a system for estimating non-rigid deformation of point cloud sequences without using any learning based approaches.

Python

PUBLICATIONS

- F. Tasyaran, K. Yildirim, K. Kaya, and M. K. Tas, "One table to count them all: Parallel frequency estimation on single-board computers," in *Euro-Par*, 2019
- S. Atito, B. A. Yanikoglu, E. Aptoula, I. Ganiyusufoglu, A. Yildiz, K. Yildirim, B. Sevilmis, and M. U. Sen, "Plant identification with deep learning ensembles," in *CLEF*, 2018